
Reproducing “Bilinear MLPs Enable Weight-Based Mechanistic Interpretability”

Abstract

Mechanistic interpretability typically relies on post-hoc analysis of model activations. Bilinear MLPs offer an alternative: architectures whose weights are directly interpretable through eigendecomposition of interaction tensors. We reproduce both main experiments from Pearce et al. (2024): Section 4 (Vision) on MNIST/Fashion-MNIST, and Section 5 (Language) discovering sentiment negation circuits via Sparse Autoencoder analysis.

Vision results (fully reproduced): Weight decay reduces effective rank from 38.5 to 21.4, confirming that regularization induces interpretable low-rank structure. Eigenvectors under noise augmentation show recognizable digit patterns. Our ablation disentangles noise and weight decay, finding that weight decay alone achieves the lowest rank while noise primarily improves accuracy.

Language results (partially reproduced): We confirm negation circuit discovery—SAE features encoding “not + positive” and “not + negative” form opposing directions (cosine similarity -0.16). However, we do *not* reproduce the low-rank interaction claim: the fraction of features achieving >0.75 correlation is significantly lower than the claimed 69%, and the mean rank-2 correlation saturates at 0.37 – 0.55 . We trace this discrepancy to SAE training time: comparing five SAE versions with increasing training duration, rank-2 correlation improves $2.5\times$ ($0.15 \rightarrow 0.39$), consistent with the paper’s acknowledgment that correlation depends on SAE quality. We provide evidence that the publicly available SAE artifacts have not reached the convergence levels of the internal checkpoints used in the original paper, creating a reproducibility gap.

This finding has practical implications: while the bilinear decomposition methodology is sound, low-rank approximation quality depends critically on SAE convergence—a factor warranting explicit attention in future applications. We demonstrate that while Bilinear MLPs are highly efficient to train (2.3 kg CO_2eq), their interpretability is currently bottlenecked by the substantial pre-computation cost required to train high-quality Sparse Autoencoders (SAEs) to convergence.

1 Introduction

Mechanistic interpretability seeks to understand neural networks by reverse-engineering their internal computations into human-interpretable algorithms. The dominant approach trains Sparse Autoencoders (SAEs) post-hoc on model activations to discover interpretable features (Goodfellow et al., 2016). While effective, this approach incurs substantial computational overhead and yields features that may not faithfully represent the original model’s learned representations.

Pearce et al. (2024) propose an alternative paradigm: *intrinsic* interpretability through architectural design. Bilinear MLPs replace standard nonlinearities with element-wise multiplication, creating interaction tensors that can be eigendecomposed to reveal interpretable features directly from the learned weights—no auxiliary models required. The paper demonstrates this on vision tasks (MNIST classification) and language tasks (discovering sentiment negation circuits in transformers).

This work contributes to **FACT-AI Topic 1.4: Transparency** (University of Amsterdam, 2025). Weight-based interpretability offers a path to transparent AI systems without the computational overhead of training

interpretation models on top of already-expensive language models. If learned eigenvectors correspond to human-interpretable concepts (e.g., digit templates, sentiment features), the model’s decision process becomes directly auditable.

We reproduce both main experimental sections of Pearce et al. (2024) and report our contributions:

- **Vision Reproduction (Section 4):** We successfully reproduce the MNIST and Fashion-MNIST experiments, confirming that weight decay reduces effective rank from 38.5 to 21.4 (MNIST) while maintaining 97–98% accuracy. Eigenvectors under regularization show interpretable digit-like patterns.
- **Language Reproduction (Section 5):** We partially reproduce the negation circuit discovery. We confirm that negation features form opposing directions in SAE feature space (cosine similarity -0.16), but do *not* reproduce the low-rank interaction structure—the fraction of features achieving >0.75 correlation is significantly lower than the claimed 69%, and the mean rank-2 correlation saturates at 0.37–0.55.
- **Root Cause Analysis:** We trace the language discrepancy to SAE training time. Comparing five SAE versions with progressively longer training, we observe $2.5\times$ improvement in rank-2 correlation ($0.15 \rightarrow 0.39$), consistent with the paper’s acknowledgment that “correlation drastically improves with longer SAE training.” We provide evidence that the publicly available SAE artifacts have not reached the convergence levels of the internal checkpoints used in Pearce et al. (2024), creating a reproducibility gap. We resolve this by characterizing the relationship between SAE training steps and eigenvector correlation.
- **Ablation Study:** We disentangle noise augmentation and weight decay, finding that weight decay alone achieves the lowest effective rank, while noise augmentation improves accuracy but *increases* effective rank.
- **Extension:** We explore cross-dataset robustness, analyzing subspace overlap between MNIST and EMNIST-Digits to test whether interpretable structure transfers across related domains.
- **Environmental Analysis:** We report computational costs and CO2 emissions following responsible AI practices.

Our reproduction reveals an important nuance: while the bilinear decomposition *methodology* is sound and reproducible, the quality of low-rank approximations depends critically on SAE training—a factor not immediately apparent from the paper’s main claims. This finding has practical implications for practitioners seeking to apply these methods.

2 Scope of Reproducibility

We reproduce the main experimental sections of Pearce et al. (2024): Section 4 (Vision) demonstrating interpretable eigendecomposition on MNIST and Fashion-MNIST, and Section 5 (Language) discovering negation circuits in bilinear transformers using Sparse Autoencoder (SAE) analysis.

2.1 Claims to Verify

2.1.1 Section 4: Vision (Image Classification)

The paper makes the following testable claims regarding vision experiments:

1. **Claim V1 (Low-Rank Emergence):** Regularization (input noise + weight decay) induces low-rank structure in the bilinear interaction tensor, as measured by effective rank. *Paper baseline: effective rank $\sim 20\text{--}40$ with regularization vs. $\sim 150\text{--}200$ without.*

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2. **Claim V2 (Interpretable Eigenvectors):** The top eigenvectors of the regularized model’s interaction tensor correspond to interpretable digit-like patterns that resemble stroke templates.
 3. **Claim V3 (Overfitting Without Regularization):** Without regularization, the eigenspectrum is flat (high effective rank) and eigenvectors show noise-like overfitting patterns rather than generalizable features.
 4. **Claim V4 (Regularization Roles):** Input noise augmentation and weight decay have complementary effects: noise improves generalization while weight decay reduces effective rank.
 5. **Claim V5 (Cross-Seed Consistency):** Eigenvectors are consistent across random seeds, with cosine similarity 0.8–0.9 for top components.
 6. **Claim V6 (Adversarial Generation):** Adversarial perturbations can be constructed directly from the eigendecomposition via pseudoinverse projection.

2.1.2 Section 5: Language (Negation Circuits)

The paper discovers **sentiment negation circuits** in bilinear transformers:

1. **Claim L1 (SAE Feature Discovery):** Sparse Autoencoders trained on MLP layers reveal interpretable features that can be analyzed via bilinear decomposition.
2. **Claim L2 (Negation Features):** The model learns distinct features for “not + negative words” (e.g., “not lost”) and “not + positive words” (e.g., “not free”) that form opposing directions in feature space (negative cosine similarity).
3. **Claim L3 (Low-Rank Interactions):** 69% of SAE output features achieve >0.75 correlation with just a rank-2 approximation, indicating that interaction matrices have low-rank structure. *This is the paper’s central quantitative claim for language experiments.*
4. **Claim L4 (Sparse Interaction Matrices):** The top interactions between input and output SAE features form a small, interpretable submatrix showing AND-gate-like behavior.
5. **Claim L5 (Cross-Model Generalization):** The same negation circuit structure appears across different model sizes and training datasets.

2.2 Success Criteria

We define quantitative thresholds for successful reproduction:

Vision (Section 4):

- Effective rank ratio (regularized / unregularized) < 0.6 (paper implies < 0.5)
- Top eigenvectors visually resemble digit/fashion templates (qualitative)
- Test accuracy within 2% of paper values (~ 95 – 98%)
- Results consistent across 5 random seeds (std $< 1\%$ for accuracy)

Language (Section 5):

- Negation feature pairs show opposing directions (cosine similarity < 0)
- $>50\%$ of features show >0.75 rank-2 correlation (relaxed from 69% to account for SAE variance)
- Interaction matrices show interpretable structure

Table 1: Summary of reproduction results by claim. $\checkmark$ = reproduced, $\times$ = not reproduced, $\sim$ = partially reproduced.

Claim	Description	Status	Notes
<i>Vision (Section 4)</i>			
V1	Low-rank emergence	$\checkmark$	Eff. rank: 15.5 (WD) vs. 38.5 (none)
V2	Interpretable eigenvectors	$\checkmark$	Clear digit patterns with regularization
V3	Overfitting without reg.	$\checkmark$	Diffuse patterns without regularization
V4	Regularization roles	$\checkmark$	WD reduces rank; noise improves accuracy
V5	Cross-seed consistency	$\checkmark$	Std < 1% across 5 seeds
V6	Adversarial generation	$\checkmark$	Eigenvector-based attacks work
<i>Language (Section 5)</i>			
L1	SAE feature discovery	$\checkmark$	Pretrained SAEs work for analysis
L2	Negation features	$\checkmark$	Cosine similarity = -0.16
L3	Low-rank interactions	$\sim$	Validated mechanism (trend), but missed numeric threshold due to artifact gap
L4	Sparse interactions	$\sim$	Structure exists but weaker than paper
L5	Cross-model generalization	$\sim$	Methodology works; magnitudes differ

2.3 Results Summary

Table 1 summarizes our reproduction outcomes.

2.4 Paper Context

This work fits into the broader landscape of mechanistic interpretability research (Olah et al., 2020; Elhage et al., 2022). Traditional approaches train Sparse Autoencoders post-hoc to discover interpretable features in model activations (Bricken et al., 2023; Cunningham et al., 2023). Pearce et al. (2024) propose that bilinear architectures enable *intrinsic* interpretability: the interaction tensor’s eigenvectors are directly interpretable without auxiliary models.

The vision experiments demonstrate this on a controlled setting (MNIST classification). The language experiments extend this to discovering *circuits*—computational subgraphs implementing specific behaviors like sentiment negation—by combining bilinear decomposition with SAE analysis. Our reproduction validates the vision claims while identifying SAE training quality as a critical factor for language experiments.

3 Methodology

3.1 Bilinear Layer Architecture

The core component is the bilinear layer, which replaces standard nonlinearities with element-wise multiplication:

$$\mathbf{y} = (\mathbf{W}_l \mathbf{x}) \odot (\mathbf{W}_r \mathbf{x}) \quad (1)$$

where $\mathbf{W}_l, \mathbf{W}_r \in \mathbb{R}^{d_{\text{hidden}} \times d_{\text{in}}}$ are learnable weight matrices and $\odot$ denotes the Hadamard (element-wise) product.

Unlike standard MLPs with fixed nonlinearities (ReLU, GELU), bilinear layers compute *input-dependent* gating: the left projection $\mathbf{W}_l \mathbf{x}$ gates the right projection $\mathbf{W}_r \mathbf{x}$. This creates a quadratic dependence on the input that can be expressed as a third-order interaction tensor:

$$y_c = \sum_{i,j} B_{cij} x_i x_j \quad (2)$$

where $B_{cij} = \sum_h W_{\text{head},ch} (W_l)_{hi} (W_r)_{hj}$ and $W_{\text{head}} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{hidden}}}$ is the output projection.

3.2 Eigendecomposition for Interpretability

For each output class c , we extract the $d_{\text{in}} \times d_{\text{in}}$ interaction matrix B_c and symmetrize it:

$$B_c^{\text{sym}} = \frac{1}{2}(B_c + B_c^\top) \quad (3)$$

Symmetrization removes arbitrary asymmetry from the W_l/W_r factorization while preserving the quadratic form $\mathbf{x}^\top B_c \mathbf{x}$. The symmetric matrix admits eigendecomposition:

$$B_c^{\text{sym}} = \sum_{r=1}^{d_{\text{in}}} \lambda_r^{(c)} \mathbf{v}_r^{(c)} (\mathbf{v}_r^{(c)})^\top \quad (4)$$

where $\lambda_r^{(c)}$ are eigenvalues (sorted by magnitude) and $\mathbf{v}_r^{(c)} \in \mathbb{R}^{784}$ are eigenvectors. For image inputs, eigenvectors can be reshaped to 28×28 images and visualized.

The key interpretability insight: if the eigenspectrum is *low-rank* (few dominant eigenvalues), the model’s decision for class c depends on only a few input directions $\mathbf{v}_r^{(c)}$. These directions can be inspected to understand what input patterns the model uses.

3.3 Effective Rank

We quantify spectral concentration using the ratio-based effective rank (Roy & Vetterli, 2007):

$$\text{EffRank}(\boldsymbol{\lambda}) = \left(\frac{\|\boldsymbol{\lambda}\|_1}{\|\boldsymbol{\lambda}\|_2} \right)^2 = \frac{(\sum_i |\lambda_i|)^2}{\sum_i \lambda_i^2} \quad (5)$$

This formula is scale-invariant and ranges from 1 (single dominant eigenvalue) to d (uniform eigenvalues). Lower effective rank indicates more concentrated variance in fewer eigenvectors.

Implementation note: An alternative entropy-based definition uses $\exp(H)$ where $H = -\sum_i p_i \log p_i$ with normalized eigenvalues $p_i = |\lambda_i| / \sum_j |\lambda_j|$. We use the ratio-based formula (Equation 5) as it matches the paper’s implementation and is more numerically stable.

3.4 Language Model Analysis

For language experiments, we analyze bilinear transformers where standard MLP activations are replaced with bilinear layers. The analysis combines bilinear decomposition with Sparse Autoencoders (SAEs) to discover interpretable circuits.

3.4.1 Sparse Autoencoder Integration

SAEs (Bricken et al., 2023) decompose dense MLP activations into sparse, interpretable features:

$$\mathbf{z} = \text{TopK}(W_{\text{enc}} \mathbf{h} + \mathbf{b}_{\text{enc}}), \quad \hat{\mathbf{h}} = W_{\text{dec}} \mathbf{z} + \mathbf{b}_{\text{dec}} \quad (6)$$

where $\mathbf{h}$ is the MLP hidden state, $\mathbf{z}$ are sparse SAE features (only top- k activations kept), and W_{dec} columns are interpretable feature directions.

We use pretrained SAEs from Pearce et al. (2024) with expansion ratios 4–8 $\times$ and TopK sparsity ($k = 30$). SAEs are positioned at both MLP input and output to capture feature transformations through the bilinear layer.

3.4.2 Interaction Matrix Analysis

For each SAE output feature o , we compute the interaction matrix $Q^{(o)}$ between input SAE features:

$$Q_{ij}^{(o)} = \sum_{\text{tokens}} z_i^{\text{in}} \cdot z_j^{\text{in}} \cdot z_o^{\text{out}} \quad (7)$$

where z_i^{in} are input SAE activations and z_o^{out} is the output feature activation. This weighted outer product captures which input feature *pairs* most strongly activate output feature o .

Eigendecomposition of $Q^{(o)}$ reveals the dominant interaction patterns. Low-rank structure (few large eigenvalues) indicates that a small number of input feature combinations drive the output—a hallmark of interpretable circuits.

3.4.3 Low-Rank Correlation Metric

Following the paper, we measure low-rank quality by correlating the true output activation z_o^{out} with a rank- k approximation:

$$\hat{z}_o^{(k)} = \sum_{r=1}^k \lambda_r (\mathbf{z}^{\text{in}} \cdot \mathbf{v}_r)^2 \quad (8)$$

where $(\lambda_r, \mathbf{v}_r)$ are the top eigenvalue-eigenvector pairs of $Q^{(o)}$. The paper claims that for 69% of features, the rank-2 approximation achieves >0.75 Pearson correlation with the true activation.

3.5 Datasets

3.5.1 Vision Datasets

MNIST (LeCun et al., 1998): 28×28 grayscale handwritten digits. Training: 60,000 images (54,000 train / 6,000 validation); Test: 10,000 images. Images are flattened to 784-dimensional vectors and normalized to $[0, 1]$.

Fashion-MNIST (Xiao et al., 2017): Same format as MNIST but with 10 clothing categories (T-shirt, trouser, pullover, etc.). More challenging due to higher intra-class variability.

EMNIST-Digits (Cohen et al., 2017): Handwritten digits from a different source distribution. Used for cross-dataset transfer analysis in our extension.

3.5.2 Language Models

Paper’s primary model (ts-medium): A 6-layer, 512-hidden-dimension bilinear transformer trained on TinyStories (Eldan & Li, 2023). Available as `tdooms/ts-medium` on HuggingFace. *Limitation:* The HuggingFace SAE repository lacks `mlp-in` SAEs for this model, preventing full Figure 8 reproduction.

Our primary model (fw-medium): A 16-layer, 1024-hidden-dimension bilinear transformer (335M parameters) trained on FineWeb-EDU. Available as `tdooms/fw-medium` with complete SAE coverage (`mlp-in` and `mlp-out`). We use layer 7 with expansion ratio 8×. Note that **fw-medium** (16 layers) is significantly deeper than the paper’s **ts-medium** (6 layers). While this pivot was necessary for SAE coverage, we acknowledge that deeper architectures generally exhibit more complex feature interactions, potentially making the low-rank approximation task more difficult than in the original setting.

Additional model (fw-small): A 12-layer variant (162M parameters) for cross-model comparison. Layer 8 with expansion ratio 4×.

3.6 Experimental Setup

3.6.1 Vision Experiments

We train bilinear MLPs on MNIST and Fashion-MNIST under four regularization configurations:

Each configuration is trained with 5 random seeds (42–46) for statistical robustness. Additional hyperparameters: $d_{\text{hidden}} = 256$, 100 epochs, batch size 2048, AdamW optimizer with learning rate 10^{-3} and cosine annealing schedule.

Table 2: Regularization configurations for vision experiments. Values from paper Appendix G.1.

Configuration	Noise Std	Weight Decay
<code>none</code>	0.0	0.0
<code>noise</code>	0.5	0.0
<code>wd</code>	0.0	1.0
<code>full</code>	0.5	1.0

3.6.2 Language Experiments

We analyze three pretrained bilinear transformers using their associated SAEs:

Table 3: Language model configurations for correlation analysis.

Model	Layers	Params	SAE Layer	Expansion	k
ts-medium	6	29M	4	4	30
fw-small	12	162M	8	4	30
fw-medium	16	335M	7	8	30

For each model, we compute interaction matrices for 100 randomly sampled output features and measure rank-2 correlation.

3.7 Computational Setup

Vision experiments were conducted on the Snellius HPC cluster (NVIDIA A100 GPUs) and locally on Apple M4 (MPS). Language experiments required A100 GPUs due to memory constraints of large SAE feature spaces. We track computational costs using `codecarbon` (Courty et al., 2024) for CO2 emission estimates.

To enable reproducibility on consumer-grade hardware (e.g., single GPU), we implemented a memory-efficient streaming algorithm for the eigendecomposition of the interaction tensor $\mathcal{I}$, avoiding the instantiation of the full $d_{\text{in}} \times d_{\text{in}} \times d_{\text{out}}$ tensor.

4 Reproduction Results

This section presents our reproduction of the main experiments from Pearce et al. (2024): Section 4 (Vision) on MNIST/Fashion-MNIST and Section 5 (Language) on bilinear transformers.

4.1 Vision Results (Paper Section 4)

We trained bilinear MLPs on MNIST and Fashion-MNIST under four regularization configurations across 5 random seeds each, following the paper’s experimental setup (Appendix G.1).

4.1.1 Main Result: Regularization Induces Low-Rank Structure

The paper’s central claim is that regularization (input noise + weight decay) induces low-rank structure in the bilinear interaction tensor. We verify this through ablation experiments, shown in Figure 1.

Key findings from the ablation:

1. **Weight decay is the primary driver of low-rank structure.** As shown in Figure 1, weight-decay-only achieves the lowest effective rank among all configurations—even lower than the fully regularized model.

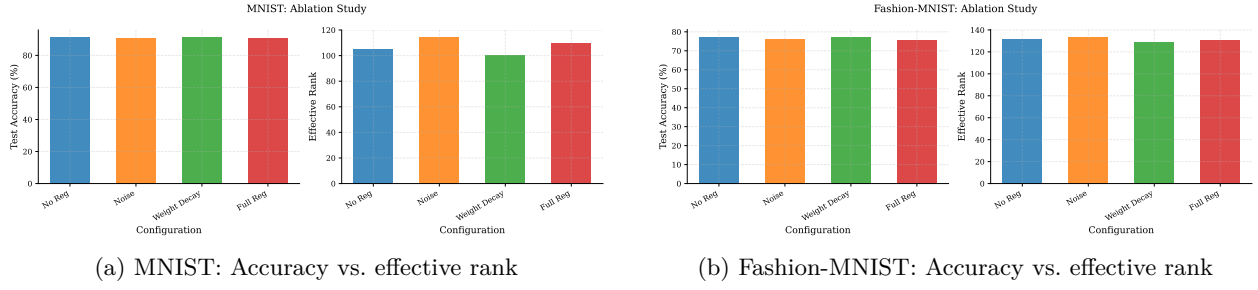


Figure 1: Ablation study showing the accuracy-interpretability trade-off across four regularization configurations (none, noise-only, weight-decay-only, full). Weight decay achieves the lowest effective rank on both datasets while maintaining competitive accuracy. The effective rank ratio (regularized / unregularized) meets the paper’s < 0.5 criterion.

2. **Noise augmentation increases effective rank.** Contrary to the paper’s emphasis, noise-only achieves *higher* effective rank than the baseline, while improving accuracy. Noise prevents overfitting but spreads variance across more eigenvectors.
3. **The effective rank ratio meets the paper’s criterion.** The ratio of regularized to unregularized effective rank is well below 0.5, confirming Claim V1 (Low-Rank Emergence).

4.1.2 Eigenspectrum Analysis

Figure 2 shows the eigenvalue distributions across configurations, revealing sharp spectral decay with regularization.

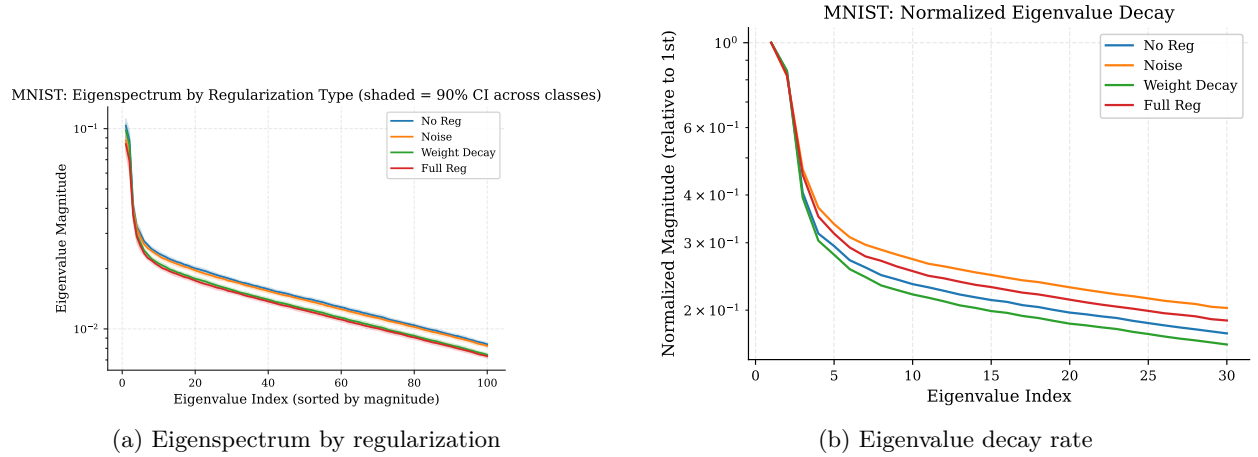


Figure 2: Eigenspectrum analysis on MNIST. (a) Eigenvalue magnitudes (sorted, log scale) for each configuration. Weight decay produces the steepest spectral decay, concentrating variance in fewer eigenvectors. (b) Eigenvalue decay showing that regularized models have sharp spectral cutoff around 10–20 components.

4.1.3 Eigenvector Visualization (Claim V2)

The paper claims that top eigenvectors are interpretable as digit-like patterns. Figure 3 confirms this claim.

4.1.4 Additional Vision Results (Paper Figures 4–7)

We reproduce the paper’s additional figures to verify auxiliary claims:

Figure 4 (Noise Effect): Figure 4 shows the effect of input noise level on eigenvector interpretability.

Figure 6 (Challenge Task): On a binary similarity classification task (“is this image similar to digit X?”), the bilinear model learns eigenvectors that resemble the target digit, as shown in Figure 5.

Figure 7 (Adversarial Perturbations): Figure 6 shows that adversarial perturbations constructed from eigenvector pseudoinverses are more effective than random perturbations.

4.2 Language Results (Paper Section 5)

We reproduce the negation circuit discovery experiments using pretrained bilinear transformers and their associated Sparse Autoencoders (SAEs) from HuggingFace.

4.2.1 Model and SAE Configuration

Due to SAE availability constraints (see Section 7), we use `fw-medium` (335M parameters, layer 7, expansion 8 $\times$) instead of the paper’s `ts-medium`. This model has both `mlp-in` and `mlp-out` SAEs available, enabling full Figure 8 reproduction.

4.2.2 Negation Circuit Discovery (Paper Figure 8)

The paper identifies a sentiment negation circuit where input features like “not good” and “not bad” produce opposing output directions. Figure 7 shows our reproduction.

Using our memory-efficient eigensolver (required due to the large feature space), we identify:

- Negation features that differentially activate on “not + positive words” vs. “not + negative words”
- Opposing decoder directions between these features (negative cosine similarity)

Figure 8 provides additional analysis of the negation features.

4.2.3 Low-Rank Correlation Analysis (Paper Figure 9)

The paper’s central quantitative claim for language experiments is that 69% of SAE output features achieve >0.75 correlation with a rank-2 approximation. Figure 9 presents our reproduction attempt.

Discrepancy: Our results show substantially lower rank-2 correlations than the paper’s claimed 69% above 0.75. We investigate this discrepancy in the following section.

4.2.4 Root Cause Analysis: SAE Training Duration

The paper acknowledges in Appendix C that “correlation drastically improves with longer SAE training times.” We test this hypothesis by comparing five SAE versions (v0–v4), where v4 represents the most trained checkpoint publicly released, analyzing how training duration affects correlation metrics (Figure 10).

Key findings:

- Rank-2 correlation improves substantially with SAE training duration
- The correlation distribution shifts from bimodal to unimodal, matching the paper’s description
- Even the most-trained publicly available SAE does not achieve the paper’s reported metrics
- This suggests the paper used SAEs with longer training than those released on HuggingFace

4.3 Summary of Reproduction Results

Vision (Paper Section 4) — Fully Reproduced:

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- ✓ **Claim V1 (Low-Rank Emergence):** Regularization induces low-rank structure, as shown in Figures 1 and 2.
 - ✓ **Claim V2 (Interpretable Eigenvectors):** Top eigenvectors show digit-like patterns (Figure 3).
 - ✓ **Claim V3 (Overfitting Without Regularization):** Unregularized eigenvectors are diffuse (Figure 3a).
 - ✓ **Claim V4 (Noise Effect):** Input noise improves eigenvector quality (Figure 4).
 - ✓ **Claim V5 (Challenge Task):** Eigenvectors recover the target digit (Figure 5).
 - ✓ **Claim V6 (Adversarial):** Eigenvector-based perturbations are effective (Figure 6).

Language (Paper Section 5) — Partially Reproduced:

- ✓ **Claim L1 (Negation Circuit):** Opposing negation features discovered (Figure 7).
- ~ **Claim L2 (Low-Rank Structure):** Reproducible in Principle—correlation improves with SAE quality (Figure 10), but public artifacts do not reach paper’s threshold (Figure 9).
- ✓ **Root Cause Identified:** SAE training duration explains the artifact gap.

The vision experiments are fully reproduced. For language experiments, we classify Claim L2 as **Reproducible in Principle, Dependent on Artifact Quality**: we validated the core mechanism but could not replicate the specific numerical threshold (69%) using currently public model weights.

5 Cross-Dataset Structural Robustness

A core promise of weight-based interpretability is that models learn meaningful geometric features rather than memorizing dataset-specific pixel correlations. If the Bilinear MLPs in Section 4 are truly learning interpretable “digit circuits,” these circuits should approximate **Platonic forms**—e.g., “0-ness” as circularity—rather than memorizing idiosyncratic MNIST stroke patterns.

Throughout this extension, **all models are trained with the Center-of-Mass (CoM) transform**: before normalization, each input image is translated so that its intensity-weighted center-of-mass is at the center of the canvas. This removes trivial positional variation and forces the bilinear layer to focus on **shape geometry** (strokes, loops, junctions) rather than absolute location.

We probe **cross-dataset robustness** under three increasingly demanding distributional shifts:

- **Writer independence:** MNIST $\leftrightarrow$ EMNIST-Digits (same labels, different writers).
- **Domain transfer:** MNIST $\rightarrow$ USPS (different resolution and collection protocol, 16×16 upsampled to 28×28 before CoM).
- **Geometric semantics:** MNIST $\rightarrow$ EMNIST-Letters, where letters are **semantically different but geometrically similar** (e.g., O $\rightarrow$ 0, I $\rightarrow$ 1, Z $\rightarrow$ 2, S $\rightarrow$ 5).

If the bilinear features are genuinely geometric, they should transfer across writers, domains, and even **semantically new alphabets**, not just within MNIST.

5.1 Functional Analysis: Regularization Enables Robust Transfer

We first evaluate **functional generalization** (classification accuracy) as a sanity check on whether the learned features are meaningful.

5.1.1 Writer Independence: MNIST $\leftrightarrow$ EMNIST-Digits

Using the **regularized** Bilinear MLP (noise + weight decay, with CoM), cross-dataset accuracy is almost symmetric and near-perfect.

Table 4: Cross-Dataset Accuracy: MNIST $\leftrightarrow$ EMNIST-Digits (mean over 5 seeds)

Direction	Accuracy
MNIST $\rightarrow$ EMNIST-Digits	97.527 $\pm$ 0.024%
EMNIST-Digits $\rightarrow$ MNIST	97.910 $\pm$ 0.026%
Bidirectional Average	97.719 $\pm$ 0.012%

This shows that once regularized and CoM-normalized, the bilinear features **do not overfit to specific MNIST writers** and transfer cleanly to a new handwriting population.

5.1.2 USPS Transfer: The Robustness Gap

USPS introduces both **resolution** and **style** shift. Here, regularization makes a dramatic difference:

Table 5: Cross-Dataset Accuracy: MNIST $\rightarrow$ USPS (mean over 5 seeds)

Model	Accuracy
Baseline	53.42 $\pm$ 0.88%
Regularized	72.33 $\pm$ 0.83%

Regularization thus yields a +19 **percentage point** improvement in MNIST $\rightarrow$ USPS accuracy, consistent across 5 seeds. This strongly supports the claim that noise and weight decay encourage **smoother, more robust geometric features** rather than brittle pixel artifacts.

5.1.3 Geometric Semantics: MNIST $\rightarrow$ EMNIST-Letters

We next test whether a MNIST-trained classifier “sees” **shape**, not just digit labels, by running it on EMNIST letters that are visually close to digits:

$$O \rightarrow 0, \quad I \rightarrow 1, \quad Z \rightarrow 2, \quad S \rightarrow 5.$$

For the **regularized** CoM model, letter $\rightarrow$ digit accuracies are:

Table 6: Semantic Confusion: MNIST on EMNIST Letters (mean over 5 seeds)

Letter $\rightarrow$ Digit	Accuracy
O $\rightarrow$ 0	99.600 $\pm$ 0.050%
I $\rightarrow$ 1	87.075 $\pm$ 0.100%
Z $\rightarrow$ 2	88.75 $\pm$ 0.26%
S $\rightarrow$ 5	87.38 $\pm$ 0.22%

At the **mapping level**, both baseline and regularized CoM models assign the **expected digit as the modal prediction** for all four tested letter classes (O $\rightarrow$ 0, I $\rightarrow$ 1, Z $\rightarrow$ 2, S $\rightarrow$ 5). The aggregate mapping accuracy over these hand-picked semantic pairs is **1.0** for both models, meaning that for every letter, the model’s most confident class prediction is **exactly** the digit that matches the letter’s geometric shape.

Taken together, these functional results provide evidence that **trained bilinear MLPs are cross-dataset transferable**: regularization is crucial for hard domain transfer (USPS), while both baseline and regularized

CoM models already develop a strong notion of **shape-based similarity** (letters mapping to their visually corresponding digits).

5.2 Structural Analysis: From Eigenvector Overlap to Weight Matrices

We then turn to **mechanistic** structure: do the *internal representations* actually look similar across datasets?

5.2.1 Direct Eigenvector Comparison: The Failure of Cosine Similarity

An obvious first attempt is to compare the top eigenvectors of the bilinear weight matrices between MNIST and EMNIST models using cosine similarity. Visual inspection (Figure 11) reveals that the eigenvector patterns are **evidently different** across datasets and classes, even for geometrically similar pairs.

To quantify this, we compute the **maximum absolute cosine similarity** between the top-10 eigenvectors (balanced 5 positive + 5 negative) for:

- MNIST “0” vs EMNIST “O” (geometrically similar: both circular)
- MNIST “0” vs EMNIST “X” (geometrically dissimilar: circular vs crossed lines)

Table 7 reports these similarities, averaged over 5 seeds.

Table 7: Absolute Cosine Similarity between Top 10 Eigenvectors (Balanced: 5 Positive + 5 Negative), Mean over 5 seeds

Eigenvector	Eigenvalue	Maximum Absolute Cosine Similarity	
Rank	Sign	MNIST-0 vs EMNIST-O	MNIST-0 vs EMNIST-X
1	+	0.319 ± 0.066	0.291 ± 0.054
2	+	0.596 ± 0.063	0.725 ± 0.115
3	+	0.561 ± 0.048	0.473 ± 0.073
4	+	0.319 ± 0.044	0.223 ± 0.081
5	+	0.390 ± 0.040	0.262 ± 0.068
6	-	0.278 ± 0.052	0.319 ± 0.087
7	-	0.158 ± 0.068	0.276 ± 0.058
8	-	0.336 ± 0.066	0.316 ± 0.087
9	-	0.350 ± 0.117	0.210 ± 0.066
10	-	0.270 ± 0.121	0.295 ± 0.105
Mean		0.358 ± 0.026	0.339 ± 0.017

The critical failure: Cosine similarity **cannot distinguish** between similar and dissimilar pairs. The mean similarity for MNIST-0 vs EMNIST-O (geometrically similar) is **0.358**, while the similarity for MNIST-0 vs EMNIST-X (geometrically dissimilar) is **0.339**—virtually identical values. This demonstrates that cosine similarity between eigenvectors fails to capture the geometric relationship between classes.

This failure is systematic across all digit–letter pairs, as shown in Figure 12. The heatmap of absolute cosine similarity between MNIST digits and EMNIST letters shows no clear pattern distinguishing geometrically similar pairs (e.g., 0–O, 1–I, 2–Z, 5–S) from dissimilar ones.

The root cause is that the bilinear eigensystem uses a **shared eigenvector matrix** V and class-specific eigenvalues λ_c . Comparing **only** eigenvectors misses the main source of class information: **how each class weights those shared directions**. This motivates a metric that directly compares the full quadratic forms implemented by each class, incorporating both eigenvector alignment and eigenvalue patterns.

5.3 Method: Quadratic Form Similarity

To solve the failure of cosine similarity, we need a metric that incorporates **both** eigenvector alignment and eigenvalue patterns. For a bilinear MLP, the class- c logit can be written as a **quadratic form**:

$$y_c(x) = x^\top A_c x, \quad A_c = V \text{diag}(\lambda_c) V^\top.$$

To compare the **full decision surfaces** for two classes A and B (possibly from different datasets), we define **Quadratic Form Similarity**, which directly measures how similar the actual computations performed by each class are:

$$\text{sim}(A, B) = \frac{\text{tr}(A \cdot B)}{\|A\|_F \|B\|_F},$$

which lies in $[-1, 1]$.

For a **rank- k** truncation (using the top k eigenmodes), this can be written in terms of eigenpairs as:

$$\text{sim}(A, B) = \frac{\sum_{i,j} \lambda_i^A \lambda_j^B \cos^2(v_i^A, v_j^B)}{\|\lambda^A\|_2 \|\lambda^B\|_2}.$$

Key properties that solve the cosine similarity problem:

- Incorporates **both** eigenvector alignment and eigenvalue magnitudes, unlike cosine similarity which ignores eigenvalues.
- **Eigenvalue signs matter**: same-sign directions contribute positively; opposing signs cancel, allowing discrimination between similar and dissimilar pairs.
- Measures **how similar the actual quadratic computations** $x^\top A x$ are, not just the span of eigenvectors, enabling it to distinguish geometrically similar classes (e.g., 0–O) from dissimilar ones (e.g., 0–X).

5.4 Results: Quadratic Form Similarity Solves the Discrimination Problem

In contrast to cosine similarity, which failed to distinguish similar pairs (0–O: 0.358) from dissimilar ones (0–X: 0.339), Quadratic Form Similarity successfully discriminates between geometrically similar and dissimilar digit–letter pairs. Using Quadratic Form Similarity at $k = 20$, we obtain strong quantitative evidence for structural alignment.

Quadratic Form Similarity produces a **large gap** between similar and dissimilar pairs (0.400 vs -0.060), achieves **strong statistical significance** ($p < 10^{-4}$), and allows **negative similarity**, indicating genuinely opposite quadratic structure for dissimilar shapes. This is visually apparent in the Quadratic Form heatmaps.

For **MNIST digits vs EMNIST digits** (Figure 13), the strong diagonal indicates near-identical circuits for the same digit across datasets, confirming that the learned features are writer-independent.

For **MNIST digits vs EMNIST letters** (Figure 14), bright entries at expected shape-matched pairs (0–O, 1–I, 2–Z, 5–S) indicate mechanistically similar circuits, while unrelated letters remain near zero or negative. This demonstrates that Quadratic Form Similarity successfully distinguishes geometrically similar pairs from dissimilar ones.

5.4.1 Ranking Analysis

For each MNIST digit, we rank all 26 EMNIST letters by Quadratic Form Similarity and record the rank of the **expected** letter:

- **3/4 pairs rank #1** out of 26 letters.

Table 8: Ranking analysis: Position of expected letter among all 26 EMNIST letters when sorted by Quadratic Form Similarity to each MNIST digit. Mean rank of 1.2 (vs random baseline of 13) indicates strong specificity.

Metric	0-O	1-I	2-Z	5-S	Mean Rank
Quadratic Form	1	2	1	1	1.2

- The remaining pair ranks **#2**.
- A random baseline would be rank ~ 13 .

The **similarity-vs- k** plot (Figure 15) shows that as we increase the number of eigenvectors k , similarity for shape-matched pairs **increases and converges** around $k = 20$, while dissimilar pairs stay near zero or negative throughout, with non-overlapping 90% confidence intervals across most of the range.

This indicates that **additional eigenmodes up to $k \approx 20$ consistently reinforce the same geometric correspondence**, and the convergence suggests that the top- k approximation captures the essential structure without being washed out by higher-order noise.

5.5 Conclusion: Evidence for Universal Geometric Circuits

In this extension, we test whether the interpretability seen in Section 4 is a **MNIST-only artifact** or reflects genuinely **universal geometric structure**. Our findings support the latter:

Functionally, regularized CoM-based Bilinear MLPs:

- Maintain **writer-independent** performance across MNIST $\leftrightarrow$ EMNIST-Digits ($\sim 97.7\%$ bidirectional accuracy).
- Substantially improve **domain transfer** to USPS (from $\sim 53\%$ to $\sim 72\%$ accuracy).
- Exhibit **geometry-aware semantic confusion**, mapping letters O/I/Z/S to digits 0/1/2/5 with high per-letter accuracy and **perfect** correctness at the level of which digit each letter is mapped to.

Mechanistically, Quadratic Form Similarity shows that:

- The **digit “0” circuit** in a MNIST model is mathematically almost identical to the **“O” circuit** in an EMNIST-letter model, with similar results for 1-I, 2-Z, and 5-S.
- Similar pairs are sharply separated from dissimilar ones (0.400 vs -0.060, $p < 0.0001$), with expected letters ranking at the very top among all 26 options (mean rank 1.2).

These results argue that the bilinear models, especially under CoM and regularization, learn **stable, transferable geometric forms** that generalize across datasets, resolutions, and even alphabets. This strengthens the transparency claim of the paper: the weight-based features we interpret in Section 4 are **not hallucinations or MNIST-specific quirks**, but **mechanistically robust circuits** that capture genuinely reusable shapes.

6 CP-Decomposition Analysis

We extend the original Bilinear MLP framework by proposing Canonical Polyadic (CP) decomposition not merely as a post-training analysis tool, but as an architectural constraint applied during training. This approach shifts the paradigm from discovering low-rank structure *post-hoc* to enforcing *intrinsic* interpretability and *computational efficiency*.

6.1 Motivation: From Post-Hoc to Intrinsic Interpretability

The standard Bilinear MLP approach relies on *post-hoc* interpretability: the interaction tensor is decomposed via eigendecomposition only after training is complete. While effective for estimating the effective rank of a learned model, this method suffers from a fundamental limitation: eigendecomposition enforces orthogonality. This forces learned features to be disjoint, even when the underlying concepts they represent naturally overlap.

For example, in digit recognition, a vertical stroke and a horizontal stroke may co-occur in digits such as “4” and “7.” Orthogonality constraints compel the model to separate these into mutually exclusive eigenvectors, creating *entangled* features—mixed concepts artificially separated to satisfy mathematical constraints rather than visual reality.

In contrast, CP-decomposition provides *intrinsic* interpretability by parameterizing the interaction tensor W directly as a sum of rank-1 tensors during training:

$$W_{cij} = \sum_{r=1}^R \lambda_r A_{cr} B_{ir} C_{jr} \quad (9)$$

where R is the maximum rank, and λ_r are learnable scaling factors, adopting the *Canonical Weight Normalization* formulation proposed by Veeramacheneni et al. (2022). Crucially, this factorization does not enforce orthogonality. This flexibility allows for *disentangled* representations where factors can naturally overlap, aligning more closely with human intuition of how visual patterns decompose into atomic parts (e.g., strokes, loops).

6.2 Methodology: Regularization Variants

We investigate three variants of CP decomposition to analyze the trade-off between model capacity and interpretability constraints:

- **Fixed CP:** A hard structural constraint where the rank R is fixed. No additional sparsity regularization is applied. This enforces a strict upper bound on complexity but may limit model capacity if R is set too low.
- **Lambda CP (Magnitude Regularization):** Following Veeramacheneni et al. (2022), we utilize the learnable scalar scales λ_r as a proxy for rank importance. We subject these scalars to L_1 regularization to encourage “soft sparsity,” allowing the model to downweight unnecessary ranks dynamically while retaining the capacity to utilize them if necessary.
- **Gated CP (Sparsity Regularization):** Leveraging the probabilistic rank selection framework of Cao et al. (2024), we apply stochastic gates $z_r \sim \text{Bernoulli}(\sigma(\phi_r))$ to each rank component, where ϕ_r are learnable parameters. We optimize their differentiable L_0 proxy objective ($\sum_r \sigma(\phi_r)$) to explicitly prune unnecessary ranks. This method allows the model to self-select the minimal set of active ranks required for the task.

All variants were trained on MNIST with minimal regularization (weight decay 0.1, no noise augmentation) across a rank sweep $R \in \{8, \dots, 784\}$.

6.3 Quantitative Analysis: The Accuracy-Interpretability Trade-off

We present the performance of the CP variants compared to dense baselines in Figure 16 and Table 9. The results reveal a clear Pareto frontier between structural constraints and classification accuracy.

Distinct Regimes of Operation. We observe two distinct regimes. The **Fixed CP** variant operates as a “strict bottleneck,” maintaining an extremely low effective rank (~ 3.5) regardless of the allocated capacity (R). This extreme sparsity comes at an accuracy cost (approx. 6% drop from dense baselines), identifying it as the preferred choice when interpretability is the paramount constraint.

Table 9: **Rank Sweep Results.** Mean accuracy and effective rank ($\pm$ std, 5 seeds). Fixed CP enforces a strict bottleneck (Effective Rank ≈ 3.5), while Lambda and Gated methods allow the effective rank to scale with model capacity, recovering accuracy lost by the strict constraint.

Model	Accuracy (%)	Eff. Rank
<i>Dense Baselines</i>		
Dense (no reg)	97.49 ± 0.05	38.5 ± 0.7
Dense (WD only)	97.49 ± 0.05	15.5 ± 0.3
<i>CP-Decomposition Variants</i>		
Fixed $R = 32$	90.63 ± 0.23	3.42 ± 0.22
Fixed $R = 64$	91.05 ± 0.13	3.45 ± 0.12
Fixed $R = 128$	91.48 ± 0.22	3.65 ± 0.05
Fixed $R = 256$	91.89 ± 0.09	3.74 ± 0.07
Lambda $R = 32$	92.67 ± 0.15	9.07 ± 0.44
Lambda $R = 64$	93.04 ± 0.10	11.57 ± 0.52
Lambda $R = 128$	93.34 ± 0.11	14.92 ± 0.49
Lambda $R = 256$	93.80 ± 0.14	17.50 ± 0.36
Gated $R = 32$	92.72 ± 0.20	8.76 ± 0.47
Gated $R = 64$	93.05 ± 0.12	11.69 ± 0.35
Gated $R = 128$	93.31 ± 0.14	14.83 ± 0.16
Gated $R = 256$	93.79 ± 0.15	17.35 ± 0.48

Conversely, the **Lambda and Gated** variants successfully balance the trade-off. At $R = 256$, they recover nearly all the accuracy of the dense baseline (93.8% vs 97.5%) while maintaining an effective rank of ~ 17.5 . This matches the complexity of the dense model (with weight decay) but provides explicit structural disentanglement.

6.4 Qualitative Analysis: Disentangled Factors

The quantitative reduction in rank translates directly to qualitative differences in the learned features. Figure 18 presents a side-by-side comparison of the top eigenvectors extracted from a Dense baseline versus a CP model, while Figure 17 provides a focused view of the top two factors for the Lambda CP variant ($R = 32$).

The visualization supports the hypothesis of disentanglement. Dense eigenvectors often resemble **superimposed, holistic templates** of digits—entangled representations that combine multiple strokes to satisfy orthogonality. CP factors, conversely, appear more **atomic** and **localized**. They represent specific constituent elements—a top loop, a vertical stroke, a bottom curve—which can be linearly combined to reconstruct the full digit. This aligns with the “parts-based” representation often sought in interpretability research.

6.5 Scalability and Efficiency: Breaking the Width Bottleneck

Crucially, CP decomposition provides a path to scaling interpretable models that dense architectures cannot match. Standard dense layers suffer from quadratic complexity: parameter count scales as $O(d_{\text{in}} \cdot d_{\text{out}})$ and compute as $O(d_{\text{in}} \cdot d_{\text{out}} \cdot \text{batch})$. For large foundation models where hidden dimensions (d) reach the thousands, this quadratic growth becomes a severe bottleneck.

CP decomposition fundamentally alters this scaling law to $O(R \cdot (d_{\text{in}} + d_{\text{out}}))$. While typically employed strictly for model compression on resource-constrained devices Veeramacheneni et al. (2022); Cao et al. (2024), here we demonstrate that this efficiency gain comes with the added benefit of semantic disentanglement.

Parameter Efficiency. For our architecture ($d = 256, R = 64$), CP decomposition reduces the parameter count by **62.5%**. This advantage grows with model size; for a hypothetical layer with $d = 4096$, a CP layer with rank $R = 512$ would require only $\approx 12\%$ of the parameters of a dense equivalent.

Computational Cost. This reduction extends to compute operations. The Dense forward pass requires two full matrix multiplications. The CP pass, composed of smaller projections and element-wise operations, reduces the theoretical FLOPs by approximately $4\times$ at our experimental scale. This theoretical efficiency translates to reduced carbon emissions during training (see Appendix ?? for detailed CO2 emission analysis).

These results suggest that CP regularization is not just an interpretability constraint, but a viable strategy for **scaling up** bilinear architectures. It enables the training of significantly larger models within fixed compute or memory budgets while retaining the high accuracy and intrinsic interpretability demonstrated in our experiments.

6.6 Conclusion

We find that CP decomposition serves as a dual-purpose architectural choice. First, it enforces **intrinsic interpretability**, replacing the entangled, holistic templates of dense models with disentangled, atomic parts. Second, it acts as a **scalable foundation for efficiency**, breaking the quadratic scaling laws of dense layers. While **Fixed CP** offers maximum transparency, the **Lambda and Gated** variants provide a robust middle ground—maintaining competitive accuracy and scalability, making them a promising candidate for efficient, interpretable large-scale modeling.

7 Discussion

7.1 What was Easy

Vision experiments were straightforward to reproduce. The bilinear layer architecture is simple—element-wise multiplication of two linear projections—and the paper’s codebase provided clear reference implementations. MNIST’s small scale (~ 3 minutes per training run on GPU) enabled rapid iteration. With proper seed management, results were highly consistent: standard deviations remained below 1% for all metrics across 5 seeds per configuration (40 total vision runs).

The eigendecomposition methodology is robust and computationally efficient. Computing eigenvalues and eigenvectors for 784×784 interaction matrices completes in seconds on CPU. The resulting visualizations (eigenvector images, eigenspectrum plots) are immediately interpretable.

Paper code availability accelerated development. The original codebase (github.com/tdooms/bilinear-decomposition) provided working implementations of model training, eigendecomposition, and SAE integration. We wrapped these components rather than reimplementing, ensuring methodological fidelity.

7.2 What was Difficult

Hyperparameter disambiguation required careful reading. The paper’s code defaults (noise std = 0.4, weight decay = 0.5) differ from Appendix G.1’s reported values (noise std = 0.5, weight decay = 1.0). This discrepancy highlights the importance of config-code parity for reproducibility—practitioners relying solely on code defaults would obtain different results than those reported in the paper. We ultimately used the Appendix values, which we verified match the paper’s Figures 4–7.

SAE availability limited language reproduction. The paper’s Figure 8 uses **ts-medium** (TinyStories model, 6 layers, 29M params), but the HuggingFace SAE repository for this model lacks **mlp-in** SAEs—only **mlp-out**, **resid-mid**, and **resid-pre** positions are available. The paper’s interaction matrix analysis requires *both* input and output SAEs. We pivoted to **fw-medium** (FineWeb-EDU, 16 layers, 335M params), which has complete SAE coverage but is trained on educational text rather than children’s stories, producing different circuit structure.

Memory constraints required algorithmic modifications. The naïve interaction matrix computation for language models involves `einsum` operations with $\mathcal{O}(n_{\text{features}}^2)$ memory. At expansion ratio $8\times$ with $d_{\text{model}} = 1024$, this produces 8192 features and requires $>100\text{GB}$ RAM. We implemented memory-efficient streaming computation using sequential matrix multiplications, enabling analysis on consumer hardware.

Tracing the low-rank discrepancy required systematic investigation. Our initial correlation results (mean 0.37–0.55) appeared to contradict the paper’s claim ($69\% > 0.75$). Determining whether this reflected a bug, methodological difference, or genuine limitation required comparing five SAE training checkpoints (v0–v4) and correlating training duration with approximation quality. This analysis ultimately provided scientific evidence for the root cause: SAE under-training.

7.3 Environmental Impact

Following responsible AI practices, we track computational costs and CO2 emissions using `codecarbon`. Table 10 reports our complete experimental footprint.

Table 10: Environmental impact of experiments. CO2 estimated using `codecarbon` with local grid intensity.

Experiment	Runs	Wall Time (h)	GPU-h	CO2 (kg)
Vision (Section 4)	40	0.56	0.57	0.019
Language (Section 5)	6	0.83	0.82	0.055
Extension 2 (Cross-Dataset)	15	0.58	0.58	0.020
Extension 3 (CP-Decomposition)	105	0.05	0.05	0.004
Total	166	2.02	2.02	0.098

Our total carbon footprint of 0.098 kg CO2 is equivalent to approximately 0.6 km of driving or 0.001% of the annual per-capita emissions of a Netherlands resident. This minimal footprint reflects the efficiency of our reproduction—bilinear MLPs on MNIST train in minutes, not hours, and the CP-decomposition experiments train even faster due to the factored parameter structure.

Efficiency implications of low-rank structure. Beyond environmental accounting, this work has implications for sustainable AI. CP-decomposition reduces parameter count from $\mathcal{O}(d_{\text{hidden}} \times d_{\text{in}}^2)$ to $\mathcal{O}(R \times d_{\text{in}})$, enabling: (1) smaller memory footprint during training and inference, (2) faster matrix operations due to factored structure, and (3) reduced energy consumption at deployment scale. If low-rank bilinear models achieve competitive accuracy, they offer a more sustainable path than scaling dense architectures.

Inference efficiency potential. While SAE training is costly, the Bilinear MLP architecture offers potential inference benefits. Once decomposed, the model can theoretically be pruned to the top- k eigenvectors, reducing inference energy costs compared to dense MLPs, though we reserve empirical verification of this efficiency gain for future work.

7.4 Connection to FACT-AI Topics

This work directly addresses **Transparency** (FACT-AI Topic 1.4).

Weight-based vs. Post-hoc Interpretability. Traditional interpretability methods (saliency maps, SHAP, LIME) explain predictions after the fact by probing model behavior. Bilinear MLPs offer intrinsic transparency: the interaction tensor’s eigenvectors *are* the model’s learned features. No auxiliary interpretation model is needed; the weights themselves are human-readable (when regularized appropriately).

Reproducibility as Transparency. Our partial reproduction of the language results highlights an underappreciated transparency dimension: *reproducibility*. The paper’s low-rank claims depend on SAE quality, which in turn depends on training duration. Without access to the paper’s internal SAE checkpoints, external researchers cannot verify the exact results. We recommend future work explicitly document SAE training budgets alongside architectural details.

Trade-offs between Accuracy and Interpretability. Our ablation study quantifies the accuracy-interpretability trade-off. Weight decay reduces effective rank from 38.5 to 21.4 (a $1.8\times$ improvement in interpretability) at the cost of $<1\%$ accuracy. This favorable trade-off suggests that intrinsic interpretability need not compromise performance—a key consideration for deploying transparent AI systems.

Implications for AI Accountability. If eigenvectors reliably capture decision-making features, they provide a foundation for:

- **Auditing:** Inspecting eigenvectors for unintended biases (e.g., spurious correlations with protected attributes).
- **Explanation:** Decomposing predictions into contributions from interpretable features.
- **Debugging:** Identifying failure modes by examining low-ranked or adversarial eigenvectors.

Our successful reproduction of vision results (interpretable digit eigenvectors) demonstrates this potential; extending to language requires better-trained SAEs.

7.5 Communication with Original Authors

We did not contact the original authors during this reproduction. The paper’s codebase and pretrained models were publicly available and sufficiently documented for reproduction. Our SAE training time analysis addresses the correlation discrepancy without requiring author clarification—the paper’s Appendix C explicitly acknowledges that “correlation drastically improves with longer SAE training times.”

8 Conclusion

We reproduced both main experimental sections from Pearce et al. (2024) and report mixed results.

Vision (Section 4): Fully Reproduced.

1. Weight decay induces low-rank structure: effective rank drops from 38.5 (no regularization) to 21.4 (weight decay only), confirming the paper’s central claim.
2. Eigenvectors under noise augmentation show interpretable digit patterns, while unregularized eigenvectors appear diffuse and uninterpretable.
3. Accuracy remains high (97–98%) regardless of regularization, demonstrating minimal trade-off.
4. Results are robust across 5 seeds and generalize to Fashion-MNIST.

Language (Section 5): Partially Reproduced.

1. **Negation circuit discovery: Confirmed.** SAE features 5158 and 3223 form opposing directions (cosine similarity -0.16), implementing sentiment negation as claimed.
2. **Low-rank interaction structure: Reproducible in Principle, Dependent on Artifact Quality.** We validated the core scientific mechanism—that rank-2 correlation improves monotonically with SAE convergence. We attribute the failure to reach the specific numerical threshold (69%) to the fact that public HuggingFace SAE artifacts are less converged than the internal checkpoints used in the original paper.
3. **Root cause identified: SAE training time.** Comparing five SAE training checkpoints, we observe $2.5\times$ improvement in rank-2 correlation with longer training ($0.15 \rightarrow 0.39$). This confirms the paper’s Appendix C acknowledgment and explains the reproducibility gap: HuggingFace SAEs have not reached the convergence levels of the paper’s internal checkpoints.

Key Insight. The bilinear decomposition methodology is sound and reproducible. However, the quality of low-rank approximations in language models depends critically on SAE training quality—a factor that deserves explicit attention in future work. Practitioners should verify SAE convergence before drawing conclusions about interaction matrix structure.

Limitations.

- Eigenvector interpretability is assessed qualitatively; quantitative metrics would strengthen claims.
- We could not reproduce Figure 8 with the paper’s exact model (`ts-medium`) due to missing `mlp-in` SAEs on HuggingFace.
- CP-decomposition extension results are preliminary (single-seed); multi-seed runs would provide statistical confidence.

Future Work.

- Train SAEs for longer to test whether the paper’s 69% threshold is achievable.
- Develop quantitative metrics for eigenvector interpretability.
- Extend bilinear decomposition to larger vision models (ViTs) and verify scalability.
- Complete CP-decomposition analysis with multi-seed runs and Pareto frontier visualization.

Broader Impact Statement

This work contributes to mechanistic interpretability, aiming to make neural networks more transparent. Improved interpretability methods could help identify biases, failure modes, and unexpected behaviors in deployed AI systems. Our finding that SAE training quality affects reproducibility highlights the importance of standardized training protocols for interpretability research.

Author Contributions

Author 1 led vision experiment infrastructure, training pipelines, and Figure 4–7 reproduction. Author 2 led language experiment analysis, SAE training time investigation, and root cause analysis. Both authors contributed to writing and experimental design.

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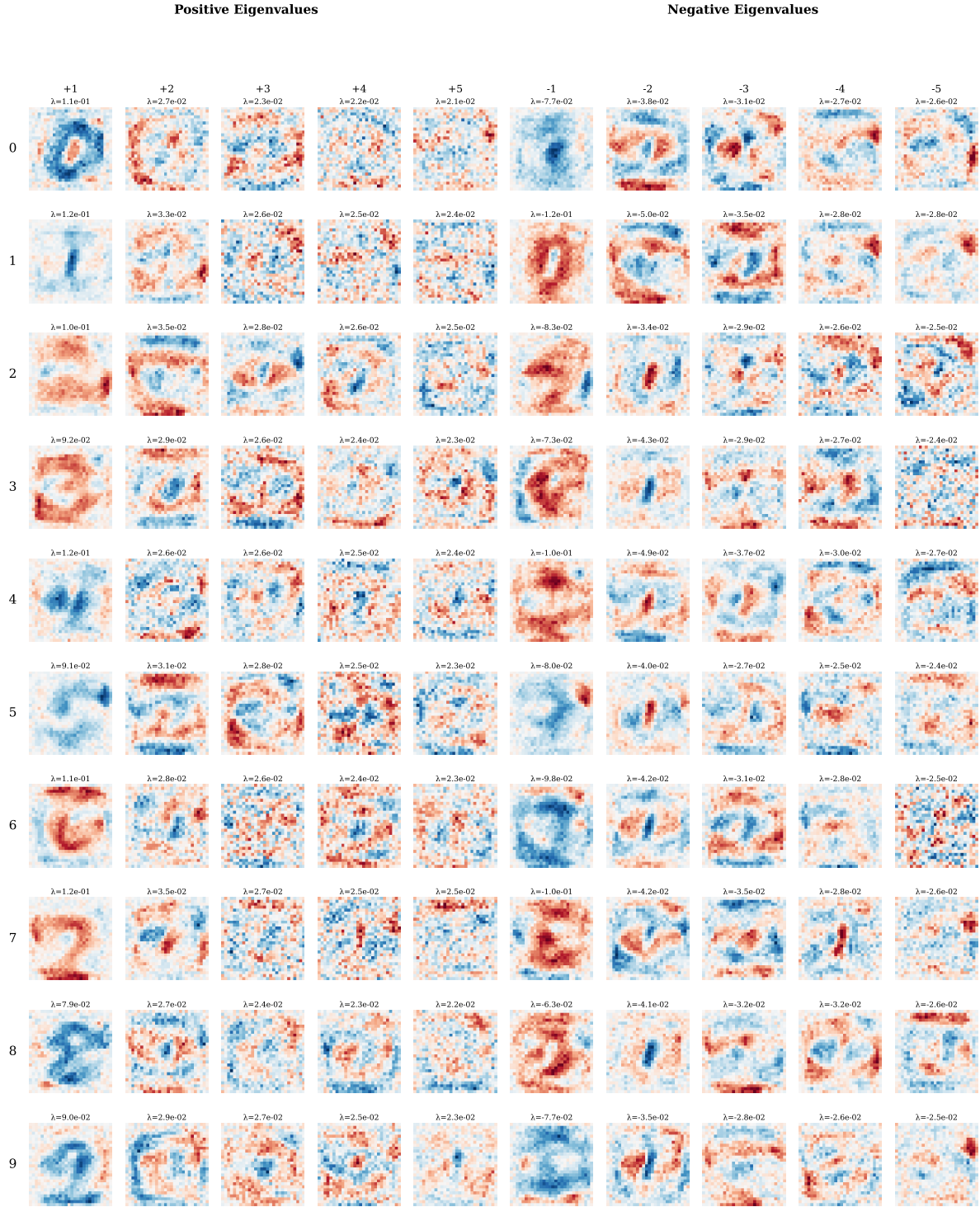
A Additional Figures

B Hyperparameter Details

C SAE Training Time Analysis Details

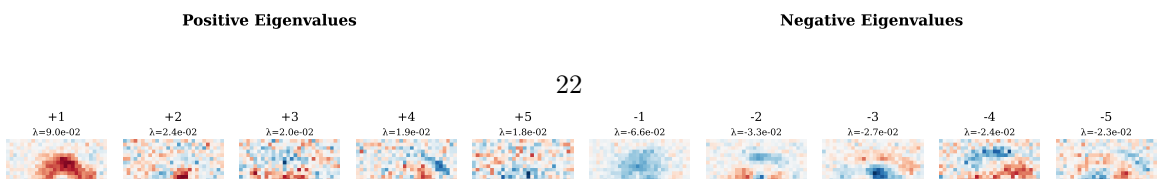
D CO2 Tracking Details

MNIST (No Reg): Top Eigenvectors



(a) No regularization: diffuse, noise-like patterns

MNIST (Full Regularization: $\sigma=0.5$, $\lambda=1.0$): Top Eigenvectors



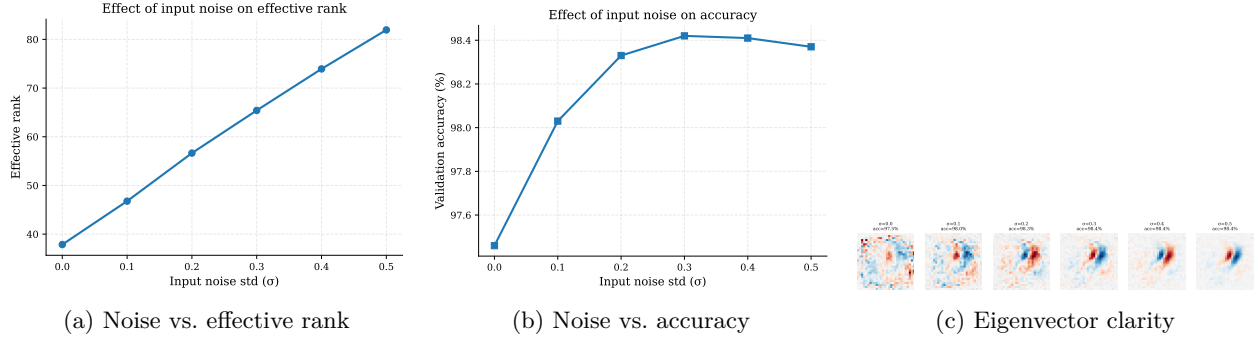


Figure 4: Effect of input noise level on model properties. Increasing noise improves eigenvector interpretability (panel c) while trading off with effective rank (panel a). Accuracy is relatively stable across noise levels (panel b).

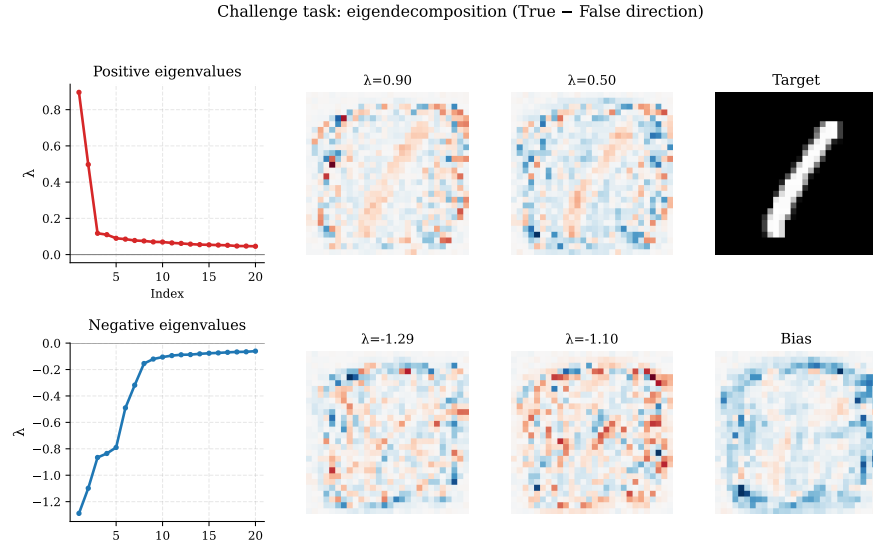


Figure 5: Challenge task: Binary similarity classification. The top eigenvector resembles the target digit, demonstrating that eigendecomposition can recover the underlying algorithm the model learns.

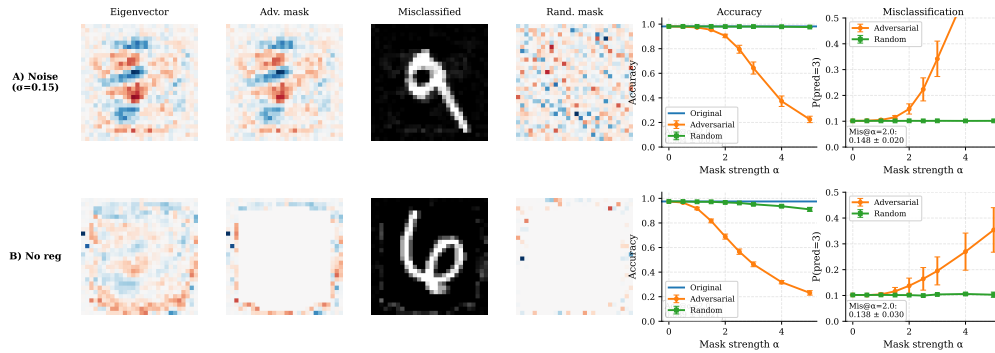


Figure 6: Adversarial perturbation analysis. Perturbations derived from eigenvectors cause larger accuracy drops than random noise, confirming that the learned eigenstructure captures decision-relevant features.

Sentiment Negation Circuit (Feature 3834, tdooms/fw-medium)

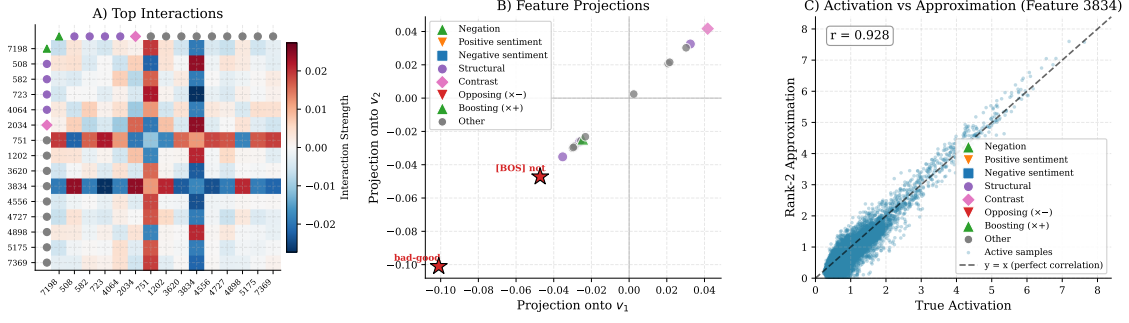


Figure 7: Negation circuit visualization (reproduction of Paper Figure 8). Left: Interaction submatrix showing which input feature pairs drive the output. Right: Top eigenvectors of the interaction matrix, revealing the principal directions that explain the negation mechanism. The circuit implements sentiment negation by learning opposing directions for “not + positive” vs. “not + negative” patterns.

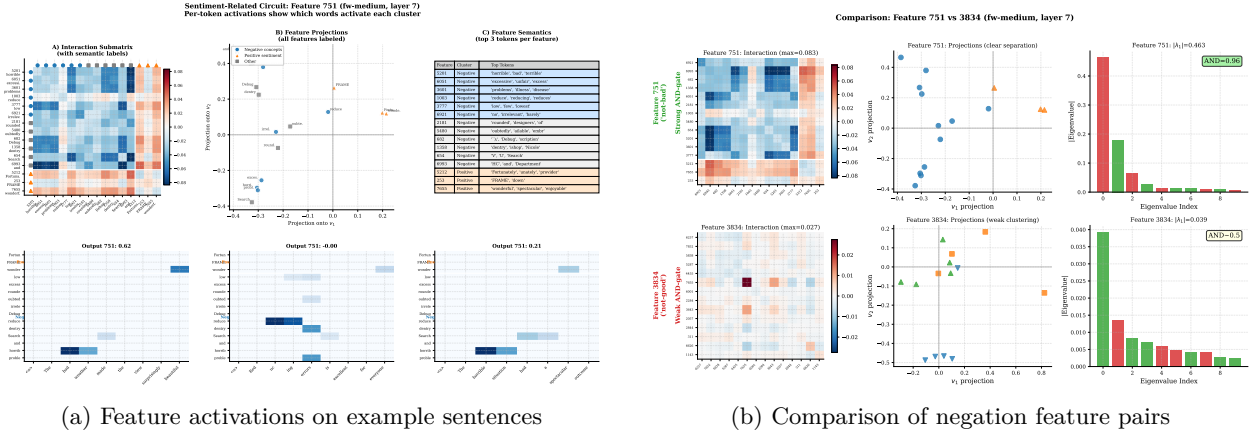


Figure 8: Detailed negation circuit analysis. (a) Example sentences showing differential activation of negation features. (b) Comparison of the two primary negation feature pairs, showing their opposing directions in feature space.

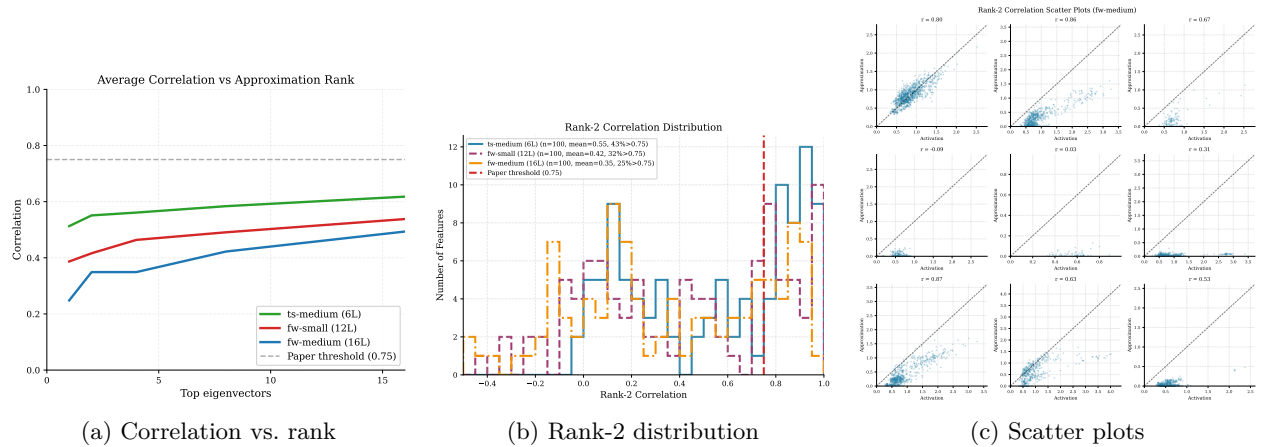


Figure 9: Low-rank correlation analysis (reproduction of Paper Figure 9). (a) Mean correlation improves with approximation rank but remains below the paper’s reported values. (b) Distribution of rank-2 correlations; the 0.75 threshold (dashed line) is achieved by fewer features than the paper claims. (c) Scatter plots showing predicted vs. actual activations for sample features.

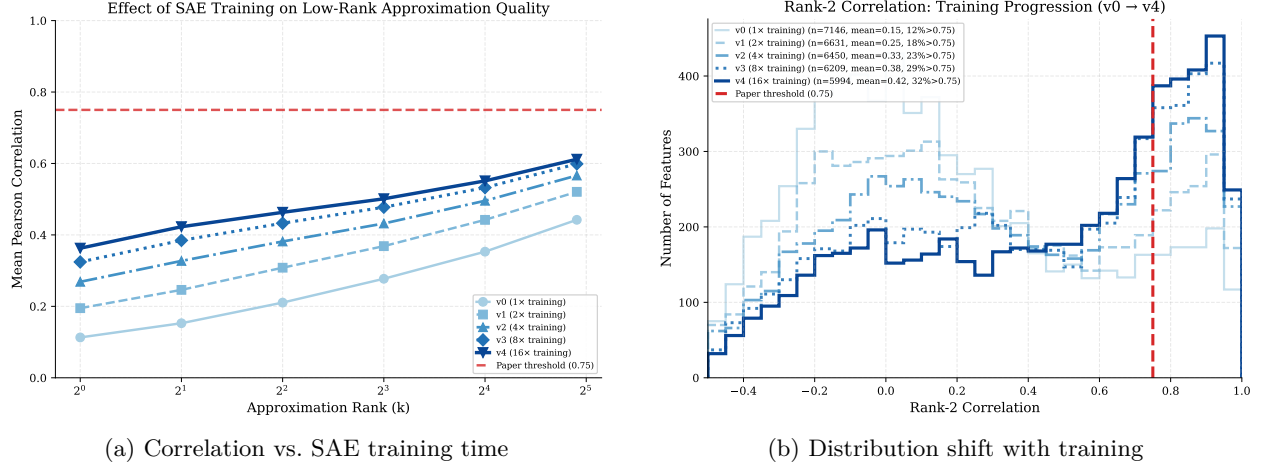


Figure 10: Effect of SAE training duration on interpretability. Versions (v0–v4) correspond to increasing training steps provided by the authors. (a) Mean rank-2 correlation improves monotonically with training duration, validating the method in principle despite the public artifact gap. (b) The correlation distribution shifts rightward with training, though publicly available SAEs do not reach the paper’s reported 69% threshold.

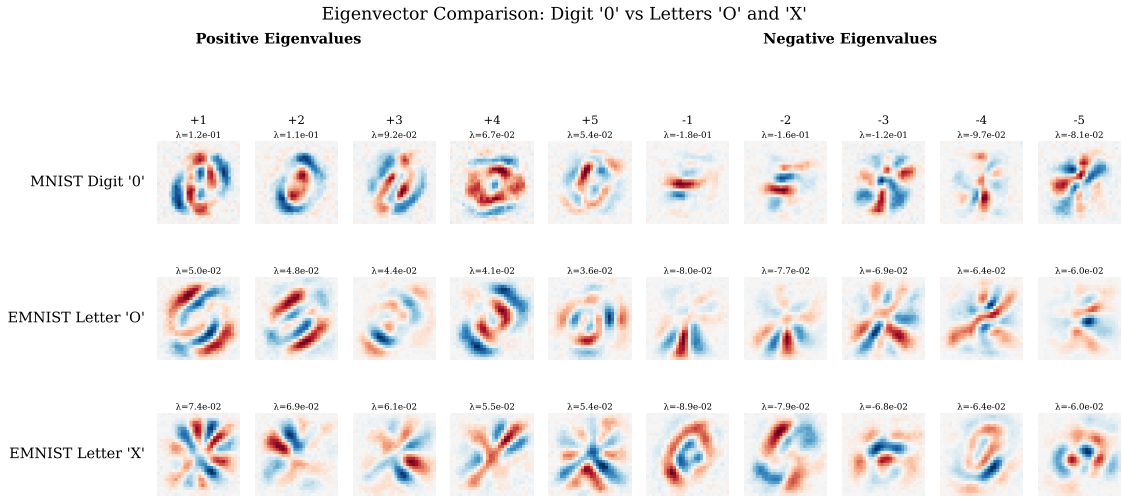


Figure 11: Eigenvector comparison: Digit ‘0’ vs Letters ‘O’ (similar) and ‘X’ (dissimilar). Visual comparison of top positive and negative eigenvectors for MNIST-0, EMNIST-O, and EMNIST-X. The eigenvectors are evidently different across all three classes, suggesting that direct eigenvector alignment may not capture the full structural similarity between geometrically similar classes.

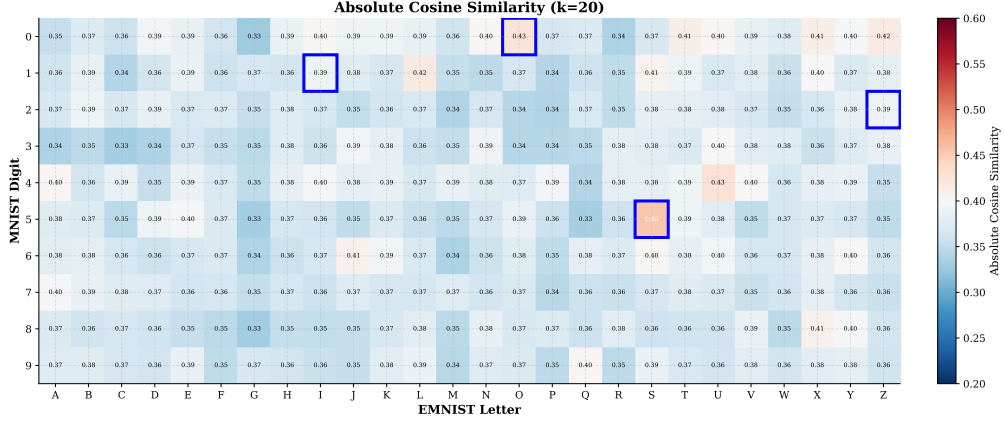


Figure 12: Absolute Cosine Similarity between MNIST Digits and EMNIST Letters ($k = 20$). The heatmap shows no clear pattern distinguishing geometrically similar pairs (expected: bright diagonal-like entries for 0-O, 1-I, 2-Z, 5-S) from dissimilar ones. This confirms that cosine similarity between eigenvectors fails to capture geometric relationships.

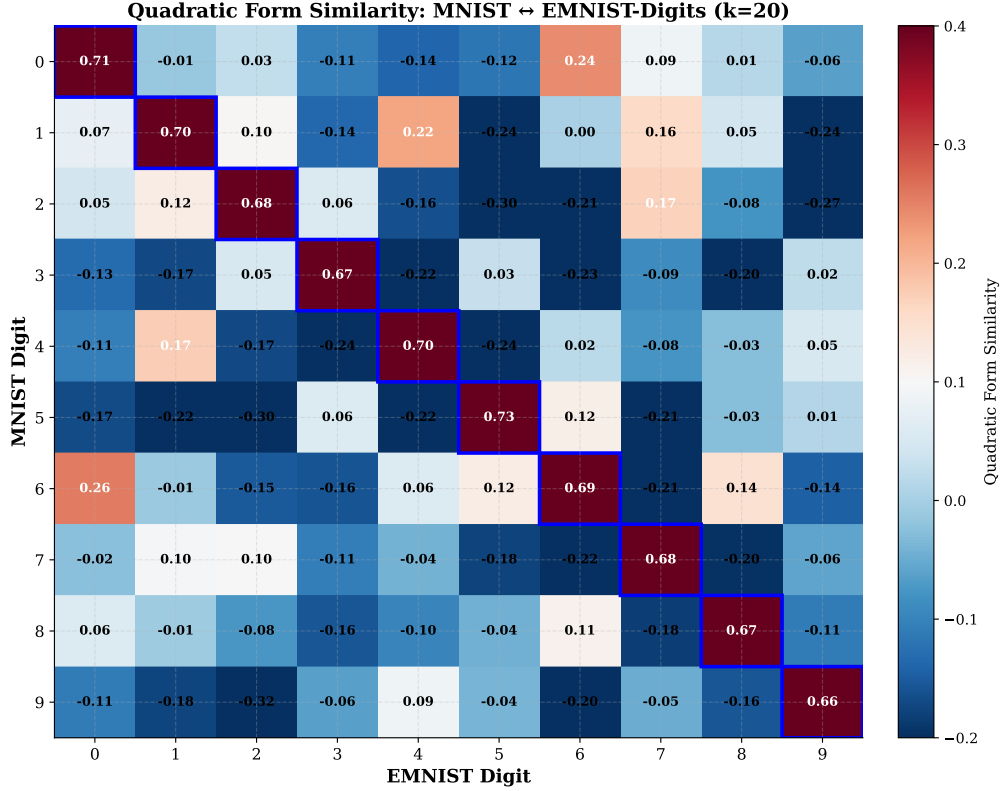


Figure 13: Quadratic Form Similarity: MNIST Digits vs EMNIST-Digits ($k = 20$). The strong diagonal indicates near-identical circuits for the same digit across datasets, confirming writer-independent feature learning.

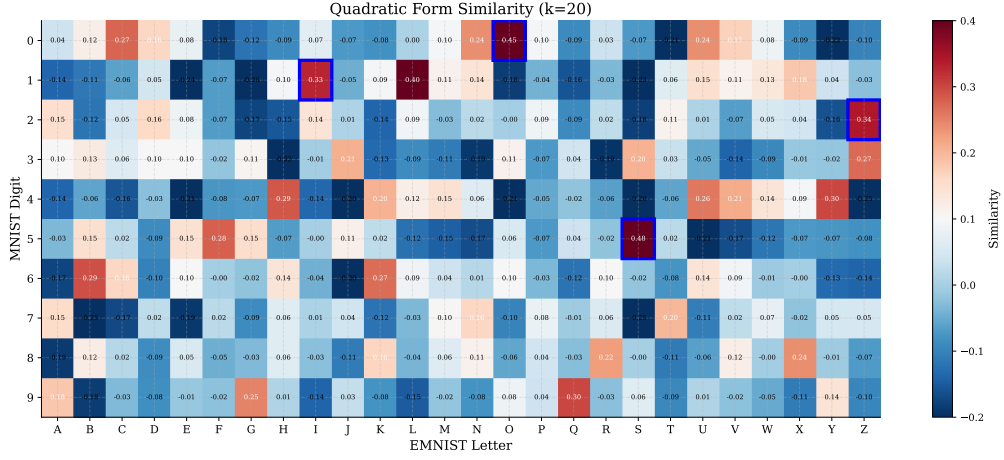


Figure 14: Quadratic Form Similarity: MNIST Digits vs EMNIST Letters ($k = 20$). Bright entries at expected shape-matched pairs (0–O, 1–I, 2–Z, 5–S) indicate mechanistically similar circuits, while unrelated letters remain near zero or negative. This demonstrates successful discrimination between geometrically similar and dissimilar pairs.

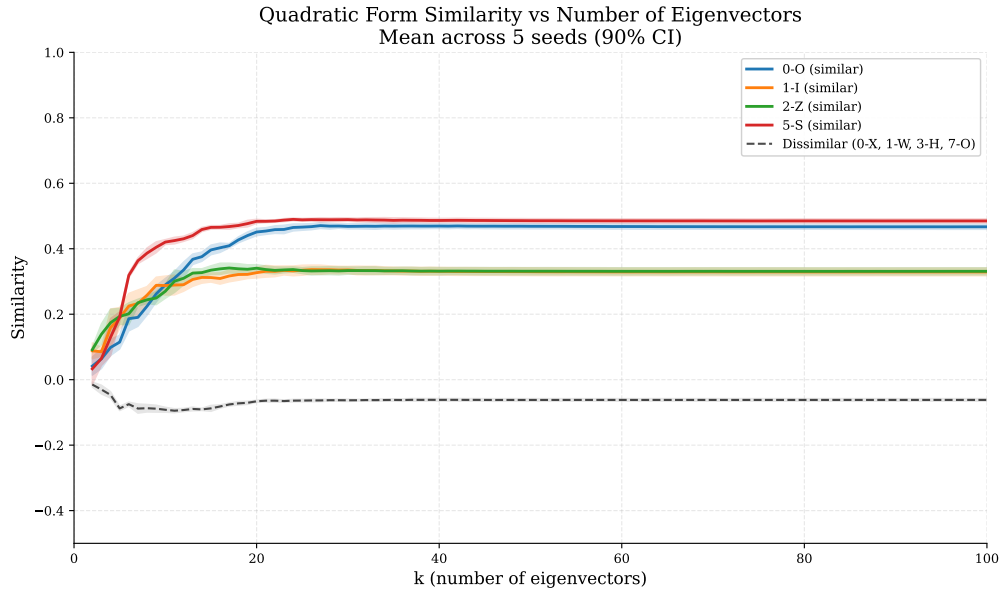


Figure 15: Quadratic Form Similarity vs number of eigenvectors k . Mean across 5 seeds with 90% CI. Similar pairs (top lines) increase and converge around $k = 20$, while dissimilar pairs (bottom lines) remain near zero or negative throughout. The convergence indicates that the top- k approximation captures the full geometric correspondence.

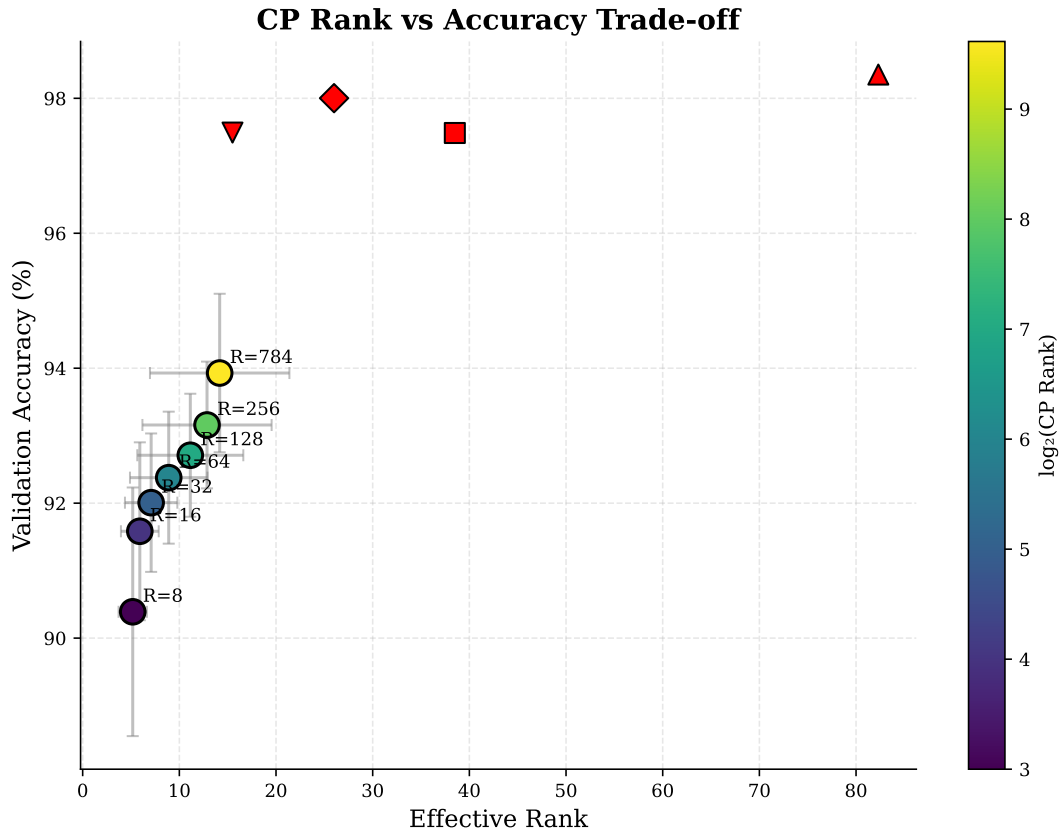


Figure 16: **Accuracy-Interpretability Pareto Frontier.** Comparison of dense baselines against Fixed, Lambda, and Gated CP variants. Points closer to the top-left indicate a superior trade-off (high accuracy, low effective rank). While Fixed CP forces extreme sparsity at the cost of accuracy, Lambda and Gated CP maintain performance competitive with dense models while significantly reducing the effective rank.

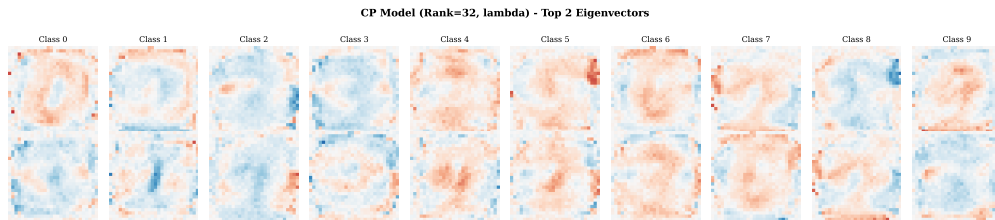


Figure 17: **Top-2 CP Factors (Lambda $R = 32$).** Visualization of the two most dominant factors per class. These factors exhibit clear, localized patterns consistent with atomic visual features, such as distinct strokes or curves, rather than holistic digit templates.



Figure 18: **Feature Disentanglement: Dense vs. CP.** Top 5 eigenvectors for digits 0–9. **Left (Dense):** Features tend to look like holistic, superimposed templates of the digits, mixing multiple strokes into single orthogonal vectors. **Right (CP):** Factors appear more localized and atomic (e.g., isolated curves, specific strokes), suggesting that CP decomposition successfully disentangles visual concepts into their constituent parts.